

1. Executive Summary & System Purpose

SkillOrbit is a full-stack peer-to-peer skill-sharing web platform built using **Python, Django, PostgreSQL, and WebSockets** [cite: 4].

The platform allows users to connect with peers based on complementary skills, schedule live 1-on-1 pair learning sessions with video calls, share video course tutorials within their active network, and communicate through instant real-time chat [cite: 4].

2. High-Level System Architecture

SkillOrbit uses a modern Model-View-Template (MVT) architecture combined with ASGI WebSockets for real-time features [cite: 4]:

SYSTEM ARCHITECTURE FLOW

```
graph LR; A[Web Browser / UI] --> B[Django App / Daphne ASGI]; B --> C[PostgreSQL & Channels]; C --> D[Jitsi Meet & YouTube API]
```

3. Comprehensive Application Breakdown

SkillOrbit is structured into **6 specialized Django applications** [cite: 4]:

3.1 Accounts Module (accounts) [cite: 4]

- **CustomUser Model:** Extends Django's user system to add custom profiles, bio, and avatar fields [cite: 4].
- **Authentication Views:** Includes user registration, secure login, and sign-out functionality [cite: 4].
- **Google OAuth Integration:** Allows users to sign in using Google credentials via Django Allauth [cite: 4].
- **Password Reset:** Provides self-service password reset using email validation links [cite: 4].
- **Clean Auth Layout:** Navigation bars are automatically hidden on authentication pages for a focused user experience [cite: 4].

3.2 Main Core Module (core) [cite: 4]

- **Master Layout (base.html):** Defines top app bar, centered main options, profile menu, and mobile bottom navigation bar [cite: 4].
- **Home Dashboard (home.html):** Displays active connections, upcoming scheduled sessions, and quick action cards [cite: 4].
- **Context Processors:** Injects connection counts, unread message badges, and pending requests into all templates [cite: 4].

3.3 Skill & Network Module (**skill**) [cite: 4]

- **Skill Directory (Skill)**: Manages the master repository of technical and creative skills [cite: 4].
- **User Skill Linking (UserSkill)**: Connects users to skills they want to teach or learn [cite: 4].
- **Connection Invites (SkillRequest)**: Handles invitations sent between users seeking to swap skills [cite: 4].
- **Active Connections (Connection)**: Stores confirmed, accepted peer relationships [cite: 4].

3.4 Scheduled Sessions Module (**session**) [cite: 4]

- **1-on-1 Session Scheduling (ScheduleSession)**: Allows users to schedule pair learning sessions with a peer [cite: 4].
- **Custom Date & Time Picker**: Features a dark-mode calendar modal compatible across all mobile and desktop browsers [cite: 4].
- **Past-Date Protection**: Prevents selecting past dates and times with instant validation warnings [cite: 4].
- **Session Lifecycle**: Pending (invitation sent), Accepted (peer accepted; activates instant Join Meeting button), Completed (session marked finished) [cite: 4].
- **Jitsi Video Integration**: Launches an instant 1-on-1 video call room inside the browser [cite: 4].

3.5 Network Video Courses Module (**course**) [cite: 4]

- **Video Course Repository (Course)**: Stores video tutorial posts created by users [cite: 4].
- **Network Connection Privacy**: Courses are visible **only to the creator and their accepted connections** [cite: 4].
- **YouTube API Integration**: Automatically extracts YouTube video IDs to render high-quality thumbnail images and embed players [cite: 4].
- **Strict Video Link Validation**: Enforces a valid YouTube URL before allowing a course to be published [cite: 4].

3.6 Real-Time Chat Module (**chat**) [cite: 4]

- **1-on-1 Conversation Rooms (Conversation)**: Pairs two connected users into a private chat channel [cite: 4].
- **Live Text Messages (Message)**: Delivers real-time chat messages using WebSockets [cite: 4].
- **Message Persistence**: Saves all chat messages to the PostgreSQL database for complete chat history [cite: 4].
- **Unread Message Badges**: Displays live unread message counts in the top navigation bar [cite: 4].

4. Complete Database Models & Relationships

SkillOrbit uses **9 database models** [cite: 4]:

CustomUser Stores user identity, authentication credentials, bio, and profile details [cite: 4].

Skill Stores master skill names (e.g. Python, Django, React, Graphic Design) [cite: 4].

UserSkill Links a user to skills they teach (teach) or skills they want to learn (learn) [cite: 4].

SkillRequest Tracks connection invitations sent from one user to another [cite: 4].

Connection Stores active, accepted friendships between two users [cite: 4].

ScheduleSession Manages scheduled 1-on-1 meetings, date/time, status, and video links [cite: 4].

Course Stores video tutorial links and descriptions uploaded by users for their connections [cite: 4].

Conversation Manages private 1-on-1 chat rooms between connected peers [cite: 4].

Message Stores individual text messages sent within a conversation [cite: 4].

5. Complete User Journey & HTML Screen Flow

Reference guide for HTML templates across the platform [cite: 4]:

Template Path	Purpose & Description
templates/base.html	Master layout with header, navigation options, profile menu, and mobile bottom bar [cite: 4].
templates/accounts/signin.html	Secure login form (navigation bars hidden) [cite: 4].
templates/accounts/signup.html	New account registration form (navigation bars hidden) [cite: 4].
templates/core/home.html	Dashboard displaying active connections, upcoming sessions, and quick stats [cite: 4].
templates/skill/explore.html	Peer discovery page to search skills and send connection requests [cite: 4].
templates/skill/connections.html	Active network page listing all connected peers with direct action buttons [cite: 4].
templates/core/sessions.html	Session hub displaying pending, accepted, and completed 1-on-1 sessions [cite: 4].
templates/session/schedule.html	Scheduling page featuring the custom dark-mode Date & Time picker [cite: 4].
templates/course/course_list.html	Video course feed displaying YouTube thumbnail cards from your network [cite: 4].
templates/course/create_course.html	Video upload form with title, description, and required YouTube link [cite: 4].
templates/chat/chat.html	Real-time 1-on-1 messaging interface with live message feed [cite: 4].

6. Key Business Rules & System Protections

- **Auth Page Layout Rules:** Top app bar, mobile bottom bar, and footer are hidden on authentication screens [cite: 4].
- **Session Past-Date Validation:** Both JavaScript frontend and Django backend validate scheduled times to prevent scheduling in the past [cite: 4].
- **Course Network Privacy:** Courses uploaded by a user are queried using `user_id__in=allowed_user_ids` (user + accepted connections) [cite: 4].
- **Required Video URL Enforcement:** Video course uploads require a valid YouTube URL before form submission is accepted [cite: 4].
- **Real-Time WebSockets Security:** Chat channels verify that the connecting user is a member of the conversation [cite: 4].

7. Installation, Setup & Deployment Guide

7.1 Local Development Setup [cite: 4]

```
# Clone and setup project
git clone https://github.com/saraths3/SkillOrbit.git
cd SkillOrbit

# Virtual environment & dependencies
python -m venv venv
source venv/bin/activate
pip install -r requirements.txt

# Migrate & Run
python SkillOrbit/manage.py migrate
python SkillOrbit/manage.py runserver
```

7.2 Production Hosting on Render.com [cite: 4]

Build Command:

```
pip install -r requirements.txt && python SkillOrbit/manage.py collectstatic --noinput &&
python SkillOrbit/manage.py migrate
```

Start Command:

```
cd SkillOrbit && daphne -b 0.0.0.0 -p $PORT SkillOrbit.asgi:application
```

8. System Status & Verification

- **System Check:** Executed `python SkillOrbit/manage.py check` — **0 issues identified** [cite: 4].
- **Static File Serving:** Configured WhiteNoise middleware for static CSS/JS delivery in production [cite: 4].